

NIRIKSHAK — AI-POWERED LEGAL METROLOGY

Principal Display Panel (PDP) Dataset & Annotation Specification Guide

1. How Many Photos Do You Need?

Because we use deep transfer learning (YOLO-OBB) combined with packaging-specific data augmentations (perspective shear, glare simulation, and rotation), you do **not** need tens of thousands of photos. Here are the exact volume targets:

Dataset Tier	Target Image Count	Expected Model Accuracy	Suitability / Use Case
Minimum Viable	150 – 250 photos	82% – 88% mAP50	Fast prototype, pipeline integration testing, and local demos.
Recommended Target (Sweet Spot)	400 – 600 photos	94% – 98% mAP50	Production-grade accuracy. High reliability across diverse retail lighting.
National / Enterprise	1,200 – 2,000 photos	>98.5% mAP50	Handles extreme edge cases (torn pouches, wet bottles, extreme blur).

Recommended Product Category Mix (For 500 Photos):

Packaging Category	Target Count	Real-World Product Examples
Rigid Cartons / Boxes	150 photos (30%)	Toothpaste boxes, soap boxes, cereal, biscuits, medicines, bulbs.
Flexible Pouches & Packets	150 photos (30%)	Chips (Lay's/Kurkure), namkeen, detergent, milk pouches, tea/coffee.
Bottles & Cans (Cylindrical)	110 photos (22%)	Cold drink cans, water bottles, shampoo bottles, oil tins, syrups.
Tubes, Jars & Blister Packs	90 photos (18%)	Face wash tubes, skin cream jars, jam jars, battery blister packs.

2. The 4 Classes to Annotate (Where Do the Boxes Go?)

You do not need to label 20 granular items. To maximize detection speed and precision, the dataset uses **4 clear classes**:

Class Name & ID	What to Box (Visual Instructions)	Why Nirikshak Needs It
Class 0: pdp_front	Draw one tight bounding box around the entire front face showing Brand Name, Product Artwork, and Net Quantity.	Mandatory under Rule 2(h) & Rule 7 . Nirikshak calculates true surface area (\$cm^2\$) and Rule 8 clearance.
Class 1: statutory_back	Draw one box around the back or information panel containing Manufacturer name, Ingredients, and Consumer Care.	Enforces Rule 6(1)(a) & 6(1)(e) (Manufacturer & customer helpline details).
Class 2: batch_stamp_area	Draw a tight box around the inkjet / laser printed batch stamp (the black dot-matrix text showing MRP, Batch No, Date).	Crucial for Rule 6(1)(d) & 6(1)(da) . Isolates faint dot-matrix print from colorful background graphics for high-accuracy OCR.

Class 3: barcode_area	Draw a tight box around the 1D EAN-13 Barcode or 2D QR code .	Used as an instant physical scale calibration anchor (\$37.29\times 25.93\$ mm) and national GS1 registry lookup.
--	--	---

+-----+ | FULL UNEDITED CAMERA PHOTO | | | |
+-----+ | | | [statutory_back] (Encloses whole information panel)
				Mfd. by: XYZ Consumer Goods India Ltd...			Ingredients: Refined Wheat Flour, Sugar, Salt...							
		Customer Care: 1800-111-222 / care@brand.in					+-----+ +-----+							
		[batch_stamp_area]		[barcode_area]				B.No: B409					MFD: 08/24	
8901314010512 | | | | MRP: Rs 50.00 (Incl.) | | | | | +-----+ +-----+ | | |
+-----+ |
+-----+

3. Package-Specific Visual Guidelines

Packaging types behave differently under 3D camera projections. Follow these specific rules:

- **Rigid Cartons (Boxes):** Draw rotated rectangles (OBB) following the 4 physical corners of the displayed panel. Do not include the top or side flaps in the front box.
- **Flexible Pouches (Chips/Namkeen):** Pouches bulge outward. Draw the box tightly around the readable central printed graphic face, ignoring the empty flat heat-sealed crimped edges at the very top and bottom.
- **Cylindrical Bottles & Cans:** Because round surfaces wrap 360°, a single photo captures only ~40% of the circumference. Box the entire *visible horizon* from the left curve edge to the right curve edge.
- **Multi-Pack / Combo Packs:** If a package contains multiple units (e.g. 'Pack of 3 Soaps'), box the outer parent wrapper as `pdp_front`.

4. Photographic Capture Do's and Don'ts

■ DO (Best Practices):

- ✓ **Capture realistic handheld angles:** ~50% flat straight-on (0°), ~50% tilted at slight handheld angles (15°–30°).
- ✓ **Include real-world store lighting:** Minor flash/ambient glare on glossy metallic laminates and soft hand/phone shadows teach the model to be robust.
- ✓ **Keep the full uncropped frame:** Leave the table surface, supermarket shelf, or fingers holding the pack. The model must learn to isolate the product from surroundings.
- ✓ **Ensure text is readable to human eyes:** Letters and numbers on the batch stamp must be sharp enough for OCR downstream.

■ DO NOT (Avoid These Costly Mistakes):

- ✗ **DO NOT pre-crop images:** Never pre-crop to the packaging boundary before labeling. The model must learn how to segment the boundary itself.
- ✗ **DO NOT label individual words:** Do not draw a box around just the word 'Maggi' or 'MRP'. The box must cover the whole `pdp_front` panel or `batch_stamp_area`.
- ✗ **DO NOT include extreme blur:** If you cannot distinguish '■50' from '■60' yourself, discard the photo.
- ✗ **DO NOT skip barcodes or stamps:** If an image shows a barcode or batch stamp, always label them. Unlabeled visible objects degrade training loss.

5. Recommended Labeling & Export Workflow

- Step 1:** Create a free project on **Roboflow (roboflow.com)** and set project type to **Oriented Bounding Box (OBB)**.
- Step 2:** Add the 4 classes: `pdp_front`, `statutory_back`, `batch_stamp_area`, `barcode_area`.
- Step 3:** Upload your 400–600 photos and label them using the rotated rectangle tool.
- Step 4:** Click **Generate Version** (leave augmentations off; our Nirikshak Python training script applies custom packaging augmentations automatically).
- Step 5:** Export as **YOLOv8 / YOLOv11 Oriented Bounding Box** format (ZIP download).
- Step 6:** Extract the images/ and labels/ folders directly into: `ComplianceEngine/stage3_pdp/training/datasets/pdp_packaging/`.
- Step 7:** Run training with one command: `python train_pdp_obb.py --epochs 50 --batch 8 --export-onnx`.